



**INSTITUTO TECNOLÓGICO
DE LAS AMERICAS**

TÍTULO DEL TRABAJO:

VPN HUB AND SPOKE PUNTO A MULTIPUNTO DMVPN FASE 3 CON IKEV2 CON
ENRUTAMIENTO DINÁMICO

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1. Objetivo de la practica

El objetivo de esta practica es configurar una VPN DMVPN Fase 3 con IKEv2 y enrutamiento dinamico EIGRP. La red implementa un router HUB y dos routers SPOKE conectados mediante un ISP simulado. La comunicacion entre las LANs se protege con IPSec/IKEv2 y se optimiza con NHRP redirect y NHRP shortcut para permitir comunicacion spoke-to-spoke directa despues de generarse el trafico.

Link del YouTube:

https://youtu.be/djtNHEr5R_k

Link de GitHub:

<https://github.com/Sael2727/DMVPN-Fase-3-con-IKEv2-y-EIGRP.git>

2. Descripcion de la topologia

La topologia utiliza un esquema hub and spoke punto a multipunto. El ISP proporciona conectividad WAN entre el HUB, SPOKE1 y SPOKE2. Cada router VPN tiene una LAN local conectada a una VPC.

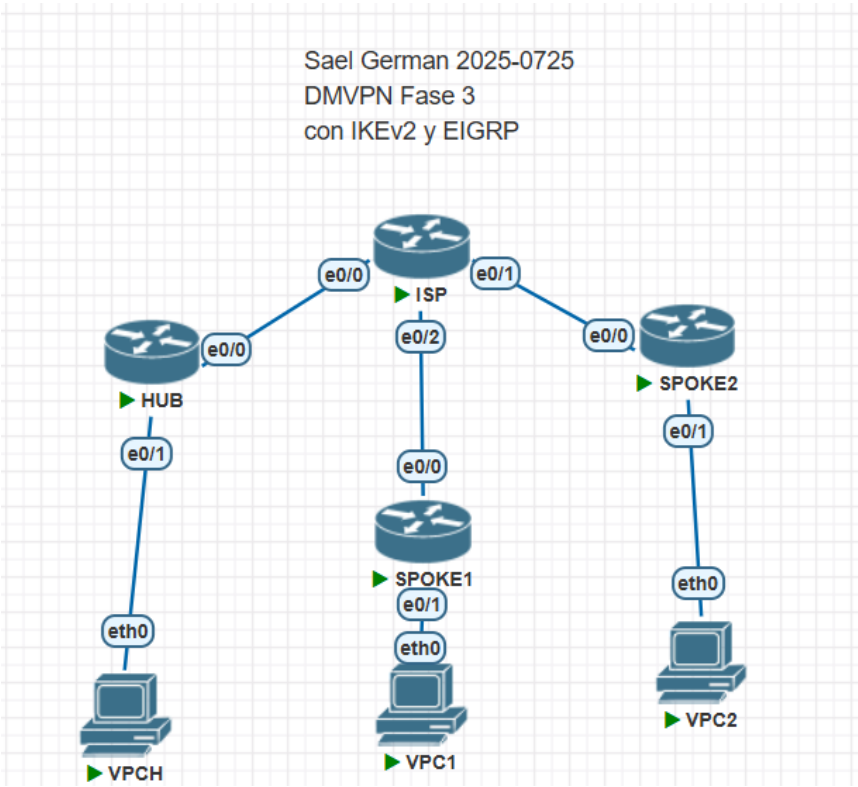


Figura 1. Topologia DMVPN Fase 3 con HUB, SPOKE1, SPOKE2, ISP y tres LANs

Equipo	Interfaz	Direccion IP	Descripcion
HUB	Ethernet0/0	10.7.25.1/30	WAN hacia ISP
ISP	Ethernet0/0	10.7.25.2/30	WAN hacia HUB
SPOKE1	Ethernet0/0	10.7.25.5/30	WAN hacia ISP
ISP	Ethernet0/2	10.7.25.6/30	WAN hacia SPOKE1
SPOKE2	Ethernet0/0	10.7.25.9/30	WAN hacia ISP

ISP	Ethernet0/1	10.7.25.10/30	WAN hacia SPOKE2
HUB	Ethernet0/1	10.7.25.65/27	LAN HUB
VPCH	eth0	10.7.25.66/27	Host LAN HUB
SPOKE1	Ethernet0/1	10.7.25.97/27	LAN SPOKE1
VPC1	eth0	10.7.25.98/27	Host LAN SPOKE1
SPOKE2	Ethernet0/1	10.7.25.129/27	LAN SPOKE2
VPC2	eth0	10.7.25.130/27	Host LAN SPOKE2
HUB	Tunnel0	10.7.25.161/27	Tunel DMVPN
SPOKE1	Tunnel0	10.7.25.162/27	Tunel DMVPN
SPOKE2	Tunnel0	10.7.25.163/27	Tunel DMVPN

3. Parametros utilizados

Parametro	Valor
Tipo de VPN	DMVPN Fase 3
Modelo	Hub and spoke punto a multipunto
Tecnologia de tunel	mGRE / NHRP
Proteccion	IPSec con IKEv2
Enrutamiento dinamico	EIGRP proceso 25
Cifrado IKEv2	AES-CBC-256
Integridad	SHA256
Diffie-Hellman	Grupo 5
Autenticacion	Pre-shared key
Clave de laboratorio	VPN12345
Transform-set IPSec	ESP-AES-256 / ESP-SHA256-HMAC
Modo IPSec	Transport
NHRP Network ID	725
NHRP Authentication	DMVPN123
Comando clave en HUB	ip nhrp redirect
Comando clave en SPOKES	ip nhrp shortcut

4. Capturas de configuracion y explicacion

En el ISP se verifican las interfaces WAN que conectan al HUB, SPOKE1 y SPOKE2. Estas interfaces permanecen en estado up/up, confirmando la conectividad fisica/logica inicial.

```
ISP#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
Ethernet0/0        10.7.25.2       YES manual up          up
Ethernet0/1        10.7.25.10      YES manual up          up
Ethernet0/2        10.7.25.6       YES manual up          up
Ethernet0/3        unassigned      YES unset  administratively down down
```

Figura 2. Verificacion de interfaces del ISP mediante show ip interface brief.

En el HUB se observan la interfaz WAN, la interfaz LAN y Tunnel0. Tunnel0 usa la IP 10.7.25.161 y aparece activa.

```
HUB#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
Ethernet0/0        10.7.25.1       YES manual up          up
Ethernet0/1        10.7.25.65      YES manual up          up
Ethernet0/2        unassigned      YES unset  administratively down down
Ethernet0/3        unassigned      YES unset  administratively down down
Tunnel0            10.7.25.161     YES manual up          up
```

Figura 3. Verificacion de interfaces del HUB mediante show ip interface brief.

En el HUB, Tunnel0 esta configurado como GRE multipoint. El comando `ip nhrp redirect` es fundamental en DMVPN Fase 3, porque permite al HUB indicar a los spokes que deben crear un camino directo cuando existe trafico spoke-to-spoke.

```
HUB#show running-config interface tunnel0
Building configuration...

Current configuration : 378 bytes
!
interface Tunnel0
  description DMVPN_PHASE3_HUB_IKEV2
  ip address 10.7.25.161 255.255.255.224
  no ip redirects
  no ip split-horizon eigrp 25
  ip nhrp authentication DMVPN123
  ip nhrp map multicast dynamic
  ip nhrp network-id 725
  ip nhrp redirect
  tunnel source Ethernet0/0
  tunnel mode gre multipoint
  tunnel key 725
  tunnel protection ipsec profile DMVPN-IKEV2-PROFILE
end
```

Figura 4. Configuración de Tunnel0 en el HUB con NHRP redirect.

En el HUB se configuraron la propuesta IKEv2, policy, keyring, profile, transform-set IPsec y profile IPsec. El perfil IPsec se aplica al tunel DMVPN para proteger el trafico GRE.

```
HUB#show running-config | section crypto
crypto ikev2 proposal IKEV2-DMVPN-PROP
  encryption aes-cbc-256
  integrity sha256
  group 5
crypto ikev2 policy IKEV2-DMVPN-POLICY
  proposal IKEV2-DMVPN-PROP
crypto ikev2 keyring IKEV2-DMVPN-KEYRING
  peer DMVPN-PEERS
    address 0.0.0.0 0.0.0.0
    pre-shared-key local VPN12345
    pre-shared-key remote VPN12345
!
crypto ikev2 profile IKEV2-DMVPN-PROFILE
  match identity remote address 0.0.0.0
  identity local address 10.7.25.1
  authentication remote pre-share
  authentication local pre-share
  keyring local IKEV2-DMVPN-KEYRING
crypto ipsec transform-set TS-DMVPN-IKEV2 esp-aes 256 esp-sha256-hmac
  mode transport
crypto ipsec profile DMVPN-IKEV2-PROFILE
  set transform-set TS-DMVPN-IKEV2
  set pfs group5
  set ikev2-profile IKEV2-DMVPN-PROFILE
```

Figura 5. Parametros criptograficos IKEv2 e IPsec en el HUB.

EIGRP proceso 25 anuncia la LAN del HUB y la red del tunel DMVPN. Tunnel0 queda como interfaz activa para formar vecindades con los spokes.

```
HUB#show running-config | section router eigrp
router eigrp 25
 network 10.7.25.64 0.0.0.31
 network 10.7.25.160 0.0.0.31
 passive-interface default
 no passive-interface Tunnel0
```

Figura 6. Configuración EIGRP en el HUB.

SPOKE1 muestra la interfaz WAN, la interfaz LAN y Tunnel0 en estado up/up.

```
SPOKE1#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
Ethernet0/0              10.7.25.5       YES manual up            up
Ethernet0/1              10.7.25.97      YES manual up            up
Ethernet0/2              unassigned      YES unset  administratively down down
Ethernet0/3              unassigned      YES unset  administratively down down
Tunnel0                  10.7.25.162     YES manual up            up
```

Figura 7. Verificación de interfaces en SPOKE1.

En SPOKE1 se configura el HUB como NHS y se habilita ip nhrp shortcut. Este comando es clave en DMVPN Fase 3 para instalar shortcuts hacia redes remotas.

```
SPOKE1#show running-config interface tunnel0
Building configuration...

Current configuration : 413 bytes
!
interface Tunnel0
 description DMVPN_PHASE3_SPOKE1_IKEV2
 ip address 10.7.25.162 255.255.255.224
 no ip redirects
 ip nhrp authentication DMVPN123
 ip nhrp map 10.7.25.161 10.7.25.1
 ip nhrp map multicast 10.7.25.1
 ip nhrp network-id 725
 ip nhrp nhs 10.7.25.161
 ip nhrp shortcut
 tunnel source Ethernet0/0
 tunnel mode gre multipoint
 tunnel key 725
 tunnel protection ipsec profile DMVPN-IKEV2-PROFILE
end
```

Figura 8. Configuración de Tunnel0 en SPOKE1 con NHRP shortcut.

SPOKE1 utiliza parametros IKEv2 equivalentes al HUB, con clave precompartida y perfil IPSec aplicado al tunel.

```

SPOKE1#show running-config | section crypto
crypto ikev2 proposal IKEV2-DMVPN-PROP
  encryption aes-cbc-256
  integrity sha256
  group 5
crypto ikev2 policy IKEV2-DMVPN-POLICY
  proposal IKEV2-DMVPN-PROP
crypto ikev2 keyring IKEV2-DMVPN-KEYRING
  peer DMVPN-PEERS
    address 0.0.0.0 0.0.0.0
    pre-shared-key local VPN12345
    pre-shared-key remote VPN12345
  !
crypto ikev2 profile IKEV2-DMVPN-PROFILE
  match identity remote address 0.0.0.0
  identity local address 10.7.25.5
  authentication remote pre-share
  authentication local pre-share
  keyring local IKEV2-DMVPN-KEYRING
crypto ipsec transform-set TS-DMVPN-IKEV2 esp-aes 256 esp-sha256-hmac
mode transport
crypto ipsec profile DMVPN-IKEV2-PROFILE
  set transform-set TS-DMVPN-IKEV2
  set pfs group5
  set ikev2-profile IKEV2-DMVPN-PROFILE

```

Figura 9. Parametros criptograficos IKEv2 e IPSec en SPOKE1.

EIGRP en SPOKE1 anuncia su LAN local 10.7.25.96/27 y la red del tunel DMVPN.

```

SPOKE1#show running-config | section router eigrp
router eigrp 25
  network 10.7.25.96 0.0.0.31
  network 10.7.25.160 0.0.0.31
  passive-interface default
  no passive-interface Tunnel0

```

Figura 10. Configuracion EIGRP en SPOKE1.

SPOKE2 muestra su interfaz WAN, LAN y Tunnel0 en estado activo.

```

SPOKE2#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	10.7.25.9	YES	manual	up	up
Ethernet0/1	10.7.25.129	YES	manual	up	up
Ethernet0/2	unassigned	YES	unset	administratively down	down
Ethernet0/3	unassigned	YES	unset	administratively down	down
Tunnel0	10.7.25.163	YES	manual	up	up

Figura 11. Verificacion de interfaces en SPOKE2.

SPOKE2 también tiene al HUB como NHS y usa ip nhrp shortcut, permitiendo que el tráfico hacia SPOKE1 cree un camino directo.

```
SPOKE2#show running-config interface tunnel0
Building configuration...

Current configuration : 413 bytes
!
interface Tunnel0
  description DMVPN_PHASE3_SPOKE2_IKEV2
  ip address 10.7.25.163 255.255.255.224
  no ip redirects
  ip nhrp authentication DMVPN123
  ip nhrp map 10.7.25.161 10.7.25.1
  ip nhrp map multicast 10.7.25.1
  ip nhrp network-id 725
  ip nhrp nhs 10.7.25.161
  ip nhrp shortcut
  tunnel source Ethernet0/0
  tunnel mode gre multipoint
  tunnel key 725
  tunnel protection ipsec profile DMVPN-IKEV2-PROFILE
end
```

Figura 12. Configuración de Tunnel0 en SPOKE2 con NHRP shortcut.

En SPOKE2 se configuraron la propuesta IKEv2, keyring, profile y el perfil IPSec DMVPN-IKEV2-PROFILE.

```

SPOKE2#show running-config | section crypto
crypto ikev2 proposal IKEV2-DMVPN-PROP
  encryption aes-cbc-256
  integrity sha256
  group 5
crypto ikev2 policy IKEV2-DMVPN-POLICY
  proposal IKEV2-DMVPN-PROP
crypto ikev2 keyring IKEV2-DMVPN-KEYRING
  peer DMVPN-PEERS
    address 0.0.0.0 0.0.0.0
    pre-shared-key local VPN12345
    pre-shared-key remote VPN12345
  !
crypto ikev2 profile IKEV2-DMVPN-PROFILE
  match identity remote address 0.0.0.0
  identity local address 10.7.25.9
  authentication remote pre-share
  authentication local pre-share
  keyring local IKEV2-DMVPN-KEYRING
crypto ipsec transform-set TS-DMVPN-IKEV2 esp-aes 256 esp-sha256-hma
  mode transport
crypto ipsec profile DMVPN-IKEV2-PROFILE
  set transform-set TS-DMVPN-IKEV2
  set pfs group5
  set ikev2-profile IKEV2-DMVPN-PROFILE

```

Figura 13. Parametros criptograficos IKEv2 e IPSec en SPOKE2.

EIGRP en SPOKE2 anuncia su LAN local 10.7.25.128/27 y la red 10.7.25.160/27 del tunel.

```

SPOKE2#show running-config | section router eigrp
router eigrp 25
  network 10.7.25.128 0.0.0.31
  network 10.7.25.160 0.0.0.31
  passive-interface default
  no passive-interface Tunnel0

```

Figura 14. Configuracion EIGRP en SPOKE2.

5. Verificacion de DMVPN, NHRP y EIGRP

El comando show dmvpn en el HUB muestra dos peers NHRP registrados, correspondientes a SPOKE1 y SPOKE2. Esto confirma que los spokes se registraron correctamente en el HUB.


```

HUB#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
    1 10.7.25.5          10.7.25.162   UP 00:14:32    D
    1 10.7.25.9          10.7.25.163   UP 00:14:01    D

```

Figura 15. Estado DMVPN en el HUB.

show ip nhrp en el HUB muestra los registros dinamicos de los spokes, relacionando las IPs de tunel con sus direcciones NBMA reales.

```

HUB#show ip nhrp
10.7.25.162/32 via 10.7.25.162
    Tunnel0 created 00:14:54, expire 01:45:05
    Type: dynamic, Flags: unique registered used nhop
    NBMA address: 10.7.25.5
10.7.25.163/32 via 10.7.25.163
    Tunnel0 created 00:14:23, expire 01:45:36
    Type: dynamic, Flags: unique registered used nhop
    NBMA address: 10.7.25.9

```

Figura 16. Tabla NHRP en el HUB.

En SPOKE1 se observa que NHRP tiene informacion de la red de SPOKE2 y su direccion NBMA. Esto evidencia el funcionamiento de NHRP shortcut en Fase 3.

```

SPOKE1#show ip nhrp
10.7.25.96/27 via 10.7.25.162
  Tunnel0 created 00:14:59, expire 01:45:00
  Type: dynamic, Flags: router unique local
  NBMA address: 10.7.25.5
  (no-socket)
10.7.25.128/27 via 10.7.25.163
  Tunnel0 created 00:14:57, expire 01:45:02
  Type: dynamic, Flags: router rib nho
  NBMA address: 10.7.25.9
10.7.25.161/32 via 10.7.25.161
  Tunnel0 created 00:16:55, never expire
  Type: static, Flags: used
  NBMA address: 10.7.25.1
10.7.25.163/32 via 10.7.25.163
  Tunnel0 created 00:14:59, expire 01:45:02
  Type: dynamic, Flags: router used nhop rib
  NBMA address: 10.7.25.9

```

Figura 17. Tabla NHRP en SPOKE1 con shortcut hacia red remota.

En SPOKE2 se observa la información NHRP hacia la red de SPOKE1, permitiendo resolución directa entre spokes.

```

SPOKE2#show ip nhrp
10.7.25.96/27 via 10.7.25.162
  Tunnel0 created 00:15:46, expire 01:44:13
  Type: dynamic, Flags: router rib nho
  NBMA address: 10.7.25.5
10.7.25.128/27 via 10.7.25.163
  Tunnel0 created 00:15:46, expire 01:44:15
  Type: dynamic, Flags: router unique local
  NBMA address: 10.7.25.9
  (no-socket)
10.7.25.161/32 via 10.7.25.161
  Tunnel0 created 00:17:11, never expire
  Type: static, Flags: used
  NBMA address: 10.7.25.1
10.7.25.162/32 via 10.7.25.162
  Tunnel0 created 00:15:46, expire 01:44:13
  Type: dynamic, Flags: router used nhop rib
  NBMA address: 10.7.25.5

```

Figura 18. Tabla NHRP en SPOKE2 con shortcut hacia red remota.

El HUB tiene vecindad EIGRP con SPOKE1 y SPOKE2 por Tunnel0.

```
HUB#show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(25)
H   Address                Interface          Hold Uptime    SRTT   RT0  Q  Seq
   (sec)                  (ms)              Cnt  Num
1   10.7.25.163             Tu0                14 00:17:49    11   1512  0   3
0   10.7.25.162             Tu0                13 00:18:22    20   1512  0   4
```

Figura 19. Vecinos EIGRP vistos desde el HUB.

SPOKE1 forma vecindad EIGRP con el HUB mediante Tunnel0.

```
SPOKE1#show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(25)
H   Address                Interface          Hold Uptime    SRTT   RT0  Q  Seq
   (sec)                  (ms)              Cnt  Num
0   10.7.25.161             Tu0                14 00:18:53    40   1470  0   6
```

Figura 20. Vecino EIGRP en SPOKE1.

SPOKE2 forma vecindad EIGRP con el HUB mediante Tunnel0.

```
SPOKE2#show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(25)
H   Address                Interface          Hold Uptime    SRTT   RT0  Q  Seq
   (sec)                  (ms)              Cnt  Num
0   10.7.25.161             Tu0                12 00:18:45    23   1470  0   6
```

Figura 21. Vecino EIGRP en SPOKE2.

6. Verificación de IKEv2 e IPsec

show crypto ikev2 sa en el HUB muestra sesiones IKEv2 en estado READY hacia los peers, confirmando negociación correcta.

```
HUB#show crypto ikev2 sa
IPv4 Crypto IKEv2 SA

Tunnel-id Local          Remote          fvrf/ivrf      Status
1          10.7.25.1/500       10.7.25.5/500  none/none      READY
   Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:5, Auth sign: PSK, Auth verify: PSK
   Life/Active Time: 86400/1213 sec

Tunnel-id Local          Remote          fvrf/ivrf      Status
2          10.7.25.1/500       10.7.25.9/500  none/none      READY
   Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:5, Auth sign: PSK, Auth verify: PSK
   Life/Active Time: 86400/1182 sec

IPv6 Crypto IKEv2 SA
```

Figura 22. Estado IKEv2 en el HUB.

SPOKE1 muestra sesiones IKEv2 en READY, incluyendo conectividad con el HUB y el spoke remoto.

```

SPOKE1#show crypto ikev2 sa
IPv4 Crypto IKEv2  SA

Tunnel-id Local Remote fvrf/ivrf Status
3 10.7.25.5/500 10.7.25.9/500 none/none READY
Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:5, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/1147 sec

Tunnel-id Local Remote fvrf/ivrf Status
1 10.7.25.5/500 10.7.25.1/500 none/none READY
Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:5, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/1257 sec

IPv6 Crypto IKEv2 SA

```

Figura 23. Estado IKEv2 en SPOKE1.

SPOKE2 muestra sesiones IKEv2 en READY, validando la negociacion segura.

```

SPOKE2#show crypto ikev2 sa
IPv4 Crypto IKEv2  SA

Tunnel-id Local Remote fvrf/ivrf Status
1 10.7.25.9/500 10.7.25.1/500 none/none READY
Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:5, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/1253 sec

Tunnel-id Local Remote fvrf/ivrf Status
3 10.7.25.9/500 10.7.25.5/500 none/none READY
Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:5, Auth sign: PSK, Auth verify: PSK
Life/Active Time: 86400/1174 sec

IPv6 Crypto IKEv2 SA

```

Figura 24. Estado IKEv2 en SPOKE2.

Los contadores pkts encaps y pkts decaps demuestran que el HUB esta cifrando y descifrando trafico GRE protegido por IPSec.

```

HUB#show crypto ipsec sa

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.7.25.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (10.7.25.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (10.7.25.9/255.255.255.255/47/0)
  current_peer 10.7.25.9 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 298, #pkts encrypt: 298, #pkts digest: 298
    #pkts decaps: 295, #pkts decrypt: 295, #pkts verify: 295
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

  local crypto endpt.: 10.7.25.1, remote crypto endpt.: 10.7.25.9
  plaintext mtu 1458, path mtu 1500, ip mtu 1500, ip mtu idb (none)
  current outbound spi: 0xE746BD30(3880172848)
  PFS (Y/N): N, DH group: none

```

Figura 25. IPSec SA en el HUB con contadores de encapsulacion y decapsulacion.

En SPOKE1 se observan paquetes encapsulados y decapsulados, lo que confirma trafico cifrado a traves de IPSec.

```

SPOKE1#show crypto ipsec sa

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.7.25.5

protected vrf: (none)
local  ident (addr/mask/prot/port): (10.7.25.5/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (10.7.25.9/255.255.255.255/47/0)
current_peer 10.7.25.9 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 44, #pkts encrypt: 44, #pkts digest: 44
  #pkts decaps: 45, #pkts decrypt: 45, #pkts verify: 45
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 10.7.25.5, remote crypto endpt.: 10.7.25.9
plaintext mtu 1458, path mtu 1500, ip mtu 1500, ip mtu idb (none)
current outbound spi: 0x388F1264(948900452)
PFS (Y/N): N, DH group: none

```

Figura 26. IPSec SA en SPOKE1.

En SPOKE2 tambien se observan contadores IPSec activos, validando el cifrado del trafico.

```

SPOKE2#show crypto ipsec sa

interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.7.25.9

protected vrf: (none)
local  ident (addr/mask/prot/port): (10.7.25.9/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (10.7.25.5/255.255.255.255/47/0)
current_peer 10.7.25.5 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 46, #pkts encrypt: 46, #pkts digest: 46
  #pkts decaps: 44, #pkts decrypt: 44, #pkts verify: 44
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 10.7.25.9, remote crypto endpt.: 10.7.25.5
plaintext mtu 1458, path mtu 1500, ip mtu 1500, ip mtu idb (none)
current outbound spi: 0xDE38E3DC(3728270300)
PFS (Y/N): N, DH group: none

```

Figura 27. IPSec SA en SPOKE2.

show crypto session muestra sesiones UP-ACTIVE en el HUB con IKEv2 SA activa e IPSec flow permitiendo GRE.

```
HUB#show crypto session
Crypto session current status

Interface: Tunnel0
Profile: IKEV2-DMVPN-PROFILE
Session status: UP-ACTIVE
Peer: 10.7.25.9 port 500
  Session ID: 2
  IKEv2 SA: local 10.7.25.1/500 remote 10.7.25.9/500 Active
  IPSEC FLOW: permit 47 host 10.7.25.1 host 10.7.25.9
    Active SAs: 2, origin: crypto map

Interface: Tunnel0
Profile: IKEV2-DMVPN-PROFILE
Session status: UP-ACTIVE
Peer: 10.7.25.5 port 500
  Session ID: 1
  IKEv2 SA: local 10.7.25.1/500 remote 10.7.25.5/500 Active
  IPSEC FLOW: permit 47 host 10.7.25.1 host 10.7.25.5
    Active SAs: 2, origin: crypto map
```

Figura 28. Sesiones crypto activas en el HUB.

SPOKE1 muestra sesiones UP-ACTIVE hacia el HUB y hacia SPOKE2, confirmando el trafico protegido.

```
SPOKE1# show crypto session
Crypto session current status

Interface: Tunnel0
Profile: IKEV2-DMVPN-PROFILE
Session status: UP-ACTIVE
Peer: 10.7.25.9 port 500
  Session ID: 1
  IKEv2 SA: local 10.7.25.5/500 remote 10.7.25.9/500 Active
  IPSEC FLOW: permit 47 host 10.7.25.5 host 10.7.25.9
    Active SAs: 2, origin: crypto map

Interface: Tunnel0
Profile: IKEV2-DMVPN-PROFILE
Session status: UP-ACTIVE
Peer: 10.7.25.1 port 500
  Session ID: 1
  IKEv2 SA: local 10.7.25.5/500 remote 10.7.25.1/500 Active
  IPSEC FLOW: permit 47 host 10.7.25.5 host 10.7.25.1
    Active SAs: 2, origin: crypto map
```

Figura 29. Sesiones crypto activas en SPOKE1.

SPOKE2 muestra sesiones UP-ACTIVE hacia el HUB y hacia SPOKE1.

```
VPCS> ping 10.7.25.66

84 bytes from 10.7.25.66 icmp_seq=1 ttl=62 time=2.249 ms
84 bytes from 10.7.25.66 icmp_seq=2 ttl=62 time=0.865 ms
^C
VPCS> trace 10.7.25.66
trace to 10.7.25.66, 8 hops max, press Ctrl+C to stop
 1  10.7.25.97    0.241 ms  0.159 ms  0.075 ms
 2  10.7.25.161   1.219 ms  0.709 ms  0.431 ms
 3  *10.7.25.66   0.628 ms (ICMP type:3, code:3, Destination port unreachable)
```

Figura 30. Sesiones crypto activas en SPOKE2.

7. Pruebas de conectividad

Desde VPC1 se realiza ping exitoso hacia VPC2. El trace muestra trafico directo por 10.7.25.163, que corresponde a Tunnel0 de SPOKE2. Esto evidencia el shortcut spoke-to-spoke.

```
SPOKE2# show crypto session
Crypto session current status

Interface: Tunnel0
Profile: IKEV2-DMVPN-PROFILE
Session status: UP-ACTIVE
Peer: 10.7.25.5 port 500
  Session ID: 2
  IKEv2 SA: local 10.7.25.9/500 remote 10.7.25.5/500 Active
  IPSEC FLOW: permit 47 host 10.7.25.9 host 10.7.25.5
    Active SAs: 2, origin: crypto map

Interface: Tunnel0
Profile: IKEV2-DMVPN-PROFILE
Session status: UP-ACTIVE
Peer: 10.7.25.1 port 500
  Session ID: 1
  IKEv2 SA: local 10.7.25.9/500 remote 10.7.25.1/500 Active
  IPSEC FLOW: permit 47 host 10.7.25.9 host 10.7.25.1
    Active SAs: 2, origin: crypto map
```

Figura 31. Prueba desde VPC1 hacia VPC2.

Desde VPC2 se realiza ping exitoso hacia VPC1. El trace muestra trafico directo por 10.7.25.162, que corresponde a Tunnel0 de SPOKE1.

```

VPCS> show ip
NAME       : VPCS[1]
IP/MASK    : 10.7.25.98/27
GATEWAY    : 10.7.25.97
DNS        :
MAC        : 00:50:79:66:68:3c
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 10.7.25.130
84 bytes from 10.7.25.130 icmp_seq=1 ttl=62 time=2.043 ms
84 bytes from 10.7.25.130 icmp_seq=2 ttl=62 time=2.756 ms
84 bytes from 10.7.25.130 icmp_seq=3 ttl=62 time=0.547 ms
^C
VPCS> trace 10.7.25.130
trace to 10.7.25.130, 8 hops max, press Ctrl+C to stop
 1  10.7.25.97    0.259 ms  0.186 ms  0.166 ms
 2  10.7.25.163   0.832 ms  0.623 ms  0.616 ms
 3  *10.7.25.130  5.078 ms (ICMP type:3, code:3, Destination port unreachable)

```

Figura 32. Prueba desde VPC2 hacia VPC1.

La prueba hacia VPCH confirma conectividad desde un spoke hacia la LAN del HUB. El mensaje Destination port unreachable al final del trace es normal en VPCS porque el trace usa UDP; el ping exitoso confirma la comunicacion.

```

VPCS> show ip
NAME       : VPCS[1]
IP/MASK    : 10.7.25.130/27
GATEWAY    : 10.7.25.129
DNS        :
MAC        : 00:50:79:66:68:3a
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 10.7.25.98
84 bytes from 10.7.25.98 icmp_seq=1 ttl=62 time=1.237 ms
84 bytes from 10.7.25.98 icmp_seq=2 ttl=62 time=0.944 ms
^C
VPCS> trace 10.7.25.98
trace to 10.7.25.98, 8 hops max, press Ctrl+C to stop
 1  10.7.25.129   0.297 ms  0.186 ms  0.286 ms
 2  10.7.25.162   0.807 ms  0.483 ms  0.388 ms
 3  *10.7.25.98   0.695 ms (ICMP type:3, code:3, Destination port unreachable)

```

Figura 33. Prueba desde un spoke hacia la LAN del HUB.

8. Resultado de la practica

La VPN DMVPN Fase 3 con IKEv2 y EIGRP fue configurada correctamente. El HUB registro a SPOKE1 y SPOKE2 mediante NHRP, EIGRP establecio vecindades y aprendio rutas dinamicamente, e IPSec/IKEv2 protegió el trafico GRE. Las pruebas de ping y traceroute demostraron conectividad entre las LANs y comunicacion spoke-to-spoke directa mediante NHRP shortcut.

El uso de ip nhrp redirect en el HUB y de ip nhrp shortcut en los spokes confirma la diferencia principal de DMVPN Fase 3 frente a Fase 2. En esta fase, el trafico puede iniciar por el HUB, pero luego NHRP instala shortcuts para optimizar la comunicacion directa entre spokes.

9. Script utilizado

```
! =====
! Sael German 2025-0725
! DMVPN Fase 3 con IKEv2 y EIGRP
! Topologia: HUB + 2 SPOKES + ISP + LANs
! Red base usada por matricula: 10.7.25.0
! =====

! 1. ISP
enable
configure terminal
hostname ISP
ip cef

interface Ethernet0/0
description WAN_HACIA_HUB
ip address 10.7.25.2 255.255.255.252
no shutdown

interface Ethernet0/1
description WAN_HACIA_SPOKE2
ip address 10.7.25.10 255.255.255.252
no shutdown

interface Ethernet0/2
description WAN_HACIA_SPOKE1
ip address 10.7.25.6 255.255.255.252
no shutdown

end
write memory

! 2. HUB
enable
configure terminal
hostname HUB
ip cef

interface Ethernet0/0
description WAN_HACIA_ISP
ip address 10.7.25.1 255.255.255.252
no shutdown

interface Ethernet0/1
description LAN_HUB
ip address 10.7.25.65 255.255.255.224
no shutdown

ip route 0.0.0.0 0.0.0.0 10.7.25.2
```

```

crypto ikev2 proposal IKEV2-DMVPN-PROP
  encryption aes-cbc-256
  integrity sha256
  group 5

crypto ikev2 policy IKEV2-DMVPN-POLICY
  proposal IKEV2-DMVPN-PROP

crypto ikev2 keyring IKEV2-DMVPN-KEYRING
  peer DMVPN-PEERS
    address 0.0.0.0 0.0.0.0
    pre-shared-key local VPN12345
    pre-shared-key remote VPN12345

crypto ikev2 profile IKEV2-DMVPN-PROFILE
  match identity remote address 0.0.0.0
  identity local address 10.7.25.1
  authentication remote pre-share
  authentication local pre-share
  keyring local IKEV2-DMVPN-KEYRING

crypto ipsec transform-set TS-DMVPN-IKEV2 esp-aes 256 esp-sha256-hmac
  mode transport

crypto ipsec profile DMVPN-IKEV2-PROFILE
  set transform-set TS-DMVPN-IKEV2
  set ikev2-profile IKEV2-DMVPN-PROFILE
  set pfs group5

interface Tunnel0
  description DMVPN_PHASE3_HUB_IKEV2
  ip address 10.7.25.161 255.255.255.224
  no ip redirects
  ip nhrp authentication DMVPN123
  ip nhrp map multicast dynamic
  ip nhrp network-id 725
  ip nhrp redirect
  no ip split-horizon eigrp 25
  tunnel source Ethernet0/0
  tunnel mode gre multipoint
  tunnel key 725
  tunnel protection ipsec profile DMVPN-IKEV2-PROFILE
  no shutdown

router eigrp 25
  no auto-summary
  passive-interface default
  no passive-interface Tunnel0
  network 10.7.25.64 0.0.0.31
  network 10.7.25.160 0.0.0.31

end
write memory

! 3. SPOKE1
enable
configure terminal
hostname SPOKE1
ip cef

interface Ethernet0/0
  description WAN_HACIA_ISP
  ip address 10.7.25.5 255.255.255.252
  no shutdown

interface Ethernet0/1

```

```

description LAN_SPOKE1
ip address 10.7.25.97 255.255.255.224
no shutdown

ip route 0.0.0.0 0.0.0.0 10.7.25.6

crypto ikev2 proposal IKEV2-DMVPN-PROP
  encryption aes-cbc-256
  integrity sha256
  group 5

crypto ikev2 policy IKEV2-DMVPN-POLICY
  proposal IKEV2-DMVPN-PROP

crypto ikev2 keyring IKEV2-DMVPN-KEYRING
  peer DMVPN-PEERS
    address 0.0.0.0 0.0.0.0
    pre-shared-key local VPN12345
    pre-shared-key remote VPN12345

crypto ikev2 profile IKEV2-DMVPN-PROFILE
  match identity remote address 0.0.0.0
  identity local address 10.7.25.5
  authentication remote pre-share
  authentication local pre-share
  keyring local IKEV2-DMVPN-KEYRING

crypto ipsec transform-set TS-DMVPN-IKEV2 esp-aes 256 esp-sha256-hmac
  mode transport

crypto ipsec profile DMVPN-IKEV2-PROFILE
  set transform-set TS-DMVPN-IKEV2
  set ikev2-profile IKEV2-DMVPN-PROFILE
  set pfs group5

interface Tunnel0
  description DMVPN_PHASE3_SPOKE1_IKEV2
  ip address 10.7.25.162 255.255.255.224
  no ip redirects
  ip nhrp authentication DMVPN123
  ip nhrp map 10.7.25.161 10.7.25.1
  ip nhrp map multicast 10.7.25.1
  ip nhrp nhs 10.7.25.161
  ip nhrp network-id 725
  ip nhrp shortcut
  tunnel source Ethernet0/0
  tunnel mode gre multipoint
  tunnel key 725
  tunnel protection ipsec profile DMVPN-IKEV2-PROFILE
  no shutdown

router eigrp 25
  no auto-summary
  passive-interface default
  no passive-interface Tunnel0
  network 10.7.25.96 0.0.0.31
  network 10.7.25.160 0.0.0.31

end
write memory

! 4. SPOKE2
enable
configure terminal
hostname SPOKE2
ip cef

```

```

interface Ethernet0/0
description WAN_HACIA_ISP
ip address 10.7.25.9 255.255.255.252
no shutdown

interface Ethernet0/1
description LAN_SPOKE2
ip address 10.7.25.129 255.255.255.224
no shutdown

ip route 0.0.0.0 0.0.0.0 10.7.25.10

crypto ikev2 proposal IKEV2-DMVPN-PROP
encryption aes-cbc-256
integrity sha256
group 5

crypto ikev2 policy IKEV2-DMVPN-POLICY
proposal IKEV2-DMVPN-PROP

crypto ikev2 keyring IKEV2-DMVPN-KEYRING
peer DMVPN-PEERS
address 0.0.0.0 0.0.0.0
pre-shared-key local VPN12345
pre-shared-key remote VPN12345

crypto ikev2 profile IKEV2-DMVPN-PROFILE
match identity remote address 0.0.0.0
identity local address 10.7.25.9
authentication remote pre-share
authentication local pre-share
keyring local IKEV2-DMVPN-KEYRING

crypto ipsec transform-set TS-DMVPN-IKEV2 esp-aes 256 esp-sha256-hmac
mode transport

crypto ipsec profile DMVPN-IKEV2-PROFILE
set transform-set TS-DMVPN-IKEV2
set ikev2-profile IKEV2-DMVPN-PROFILE
set pfs group5

interface Tunnel0
description DMVPN_PHASE3_SPOKE2_IKEV2
ip address 10.7.25.163 255.255.255.224
no ip redirects
ip nhrp authentication DMVPN123
ip nhrp map 10.7.25.161 10.7.25.1
ip nhrp map multicast 10.7.25.1
ip nhrp nhs 10.7.25.161
ip nhrp network-id 725
ip nhrp shortcut
tunnel source Ethernet0/0
tunnel mode gre multipoint
tunnel key 725
tunnel protection ipsec profile DMVPN-IKEV2-PROFILE
no shutdown

router eigrp 25
no auto-summary
passive-interface default
no passive-interface Tunnel0
network 10.7.25.128 0.0.0.31
network 10.7.25.160 0.0.0.31

end

```

```
write memory
```

```
! VPCS  
! VPCH  
ip 10.7.25.66 255.255.255.224 10.7.25.65  
save
```

```
! VPC1  
ip 10.7.25.98 255.255.255.224 10.7.25.97  
save
```

```
! VPC2  
ip 10.7.25.130 255.255.255.224 10.7.25.129  
save
```

10. Referencia

1. Reconocimiento especial: Troubleshooting, base del script y documentación apoyado en Inteligencia Artificial.